

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

Anonymous authors
Anonymous submission

ABSTRACT

Diffusion probabilistic models (DPMs) now achieve strong sample quality in image and video synthesis, yet their outputs can still violate factual regularities in ways that are visually plausible, such as producing an impossible duplicate cup beside another cup. Progress on this failure mode is limited by the absence of a practical quantitative protocol: without comparable rates, margins, and metric relationships, it is difficult to determine which sampling choices reduce hallucinations and which merely improve perceptual appearance. We therefore study a measurable subclass, **counting hallucination**, where a generated sample contains an incorrect number of instances or structured parts, such as a hand with six fingers even though the reference data contain only five-finger hands. We introduce **CountHalluSet**, a three-dataset suite—ToyShape, SimObject, and RealHand—whose counting criteria make each comparison explicit in terms of dataset, object category, and valid count range. With this suite, we define a standardized evaluation protocol and quantify how solver type, ODE solver order, sampling steps, and initial noise change counting hallucination rates under matched generation budgets. Across these controlled comparisons, the empirical patterns show that common perceptual metrics such as Fréchet Inception Distance (FID) do not consistently track counting hallucinations, motivating direct reporting of count-based failure rates alongside standard image-quality metrics.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion models, have become a high-performing generative modeling family, producing strong samples for text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). This quality, however, does not guarantee quantitative fidelity to factual properties in the training data: generated outputs can preserve perceptual realism while violating real-world constraints. The most practically visible examples are count errors in spatially intricate objects, including extra human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023), where the relevant comparison is not only image quality but whether the generated count matches the reference rule.

Recent work has made these failures more visible by cataloging and stress-testing text-guided diffusion models. For example, Borji (2023) organize qualitative shortcomings in text-to-image generation, Liu et al. (2024) use adversarial search to expose undesirable behaviors and failure cases, and Wu et al. (2023) iteratively discover and mitigate spurious correlations. These studies provide useful failure taxonomies, but they leave open the quantitative question central to this paper: under matched datasets, solvers, and metrics, how often do factual count violations occur?

The more specific notion of *hallucination* in diffusion models has recently attracted attention (Aithal et al., 2024) because it differs from both prompt-alignment errors and obvious low-quality artifacts. Prompt-based failures can often be scored with condition-output alignment measures such as CLIP score (Hessel et al., 2021) or VQA score (Huang et al., 2025; Zarei et al., 2024), and severe blur or distortion is usually apparent from perceptual quality. Hallucinated samples, by contrast, can look plausible while breaking factual constraints implied by the training distribution, including incorrect object counts, physically impossible geometries, or other violations of real-world laws. This gap between visual plausibility and factual correctness creates an evaluation problem: current AI systems

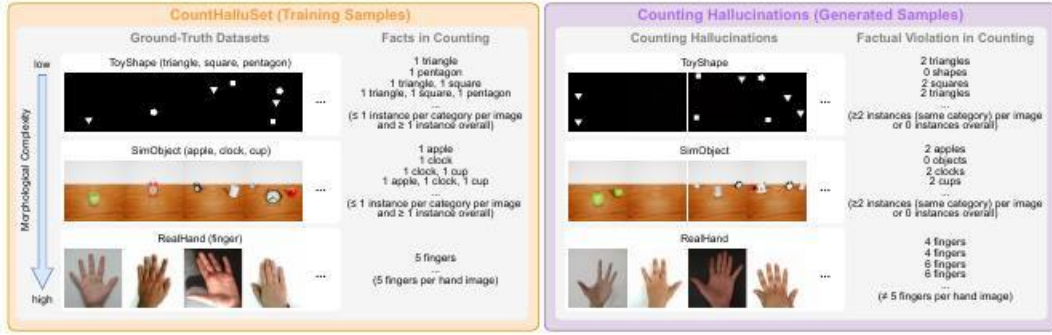


Figure 1: Side-by-side comparison between reference samples in **CountHalluSet** (left) and model-generated **counting hallucinations** (right). The comparison is defined by dataset-specific count rules: ToyShape and SimObject allow at most one instance per category and require at least one instance overall, while RealHand requires exactly five fingers. A generated image is counted as a hallucination when it violates the corresponding rule, for example by producing two apples or a normal-background image with no objects in SimObject, patterns absent from the reference dataset.

can miss the relevant distinction, and even state-of-the-art large vision-language models such as GPT-4o can struggle to count fingers in otherwise clear hand images. A standardized quantitative protocol is therefore necessary for comparing methods by factual error rates rather than by appearance alone, especially if diffusion models are to serve as world models with strict consistency requirements (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among hallucinated attributes such as color, spatial relation, geometry, count, and physical realism, count provides the clearest path to comparable empirical evidence because it can be mapped to an explicit rule and reported as a rate. We therefore define **counting hallucination** as the count-based factual error studied throughout the paper. *In this setting, the number of objects in training data is explicitly predetermined. Consequently, a generated image is deemed hallucinated whenever its object counts deviate from these predefined values.* For instance, a generated hand with six fingers is a counting hallucination relative to RealHand because the dataset contains five-finger hands, as shown in the last row of Fig. 1. To support comparisons across increasing object complexity, CountHalluSet contains ToyShape (triangle, square, pentagon), SimObject (mug, apple, clock), and RealHand (finger). Figure 1 pairs representative reference images with corresponding hallucinations so that the empirical target of each dataset is visually explicit.

The protocol turns each generated sample into quantitative evidence by first identifying whether the sample is suitable for counting and then estimating object counts with dataset-specific counting models. Using the same datasets and counting models across conditions, we compare sampling steps, initial noise, ODE solver order, and solver type by their effect on counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR). We then test whether these count-based outcomes align with standard perceptual metrics, including Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021).

The main quantitative findings are:

- Increasing ODE sampling steps from 25 to 100 reduces CHR on simpler synthetic datasets, but the same comparison often fails to reduce CHR for RealHand, showing that more numerical effort does not automatically improve count factuality.
- CHR has weak and condition-dependent correlations with FID and MUSIQ, whereas NCFR shows strong positive correlations with these perceptual metrics, separating count errors from general visual failures.
- Motivated by the lower CHR of simpler structural targets, we introduce joint-diffusion models (JDM), which reduce both CHR and NCFR relative to vanilla LDM on RealHand across the reported solver and step settings.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Motivated by the extensive literature on hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), recent diffusion-model research has begun to ask analogous questions for generated images. A key qualitative account from Aithal et al. (2024) proposes that hallucinated samples may arise from interpolation between distinct data modes. What remains missing is the quantitative counterpart: a protocol that reports hallucination rates under controlled model, solver, dataset, and metric comparisons.

This gap motivates our focus on identifying hallucinations in diffusion-generated outputs with explicit count rules and measurable rates. By holding the evaluation protocol fixed while varying sampling steps, solver order, solver type, and initialization, we can separate empirical changes in hallucination severity from broader changes in perceptual quality.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

Visual Question Answering (VQA) score (Huang et al., 2025; Zarei et al., 2024) and CLIP score (Hessel et al., 2021) are standard quantitative measures for text-guided image generation, but they evaluate condition-output alignment rather than direct agreement with dataset-level count facts. Their dependence on language-based models also introduces a measurement risk: pretrained language models have a statistical lower bound on hallucinating certain fact types, limiting detection accuracy when language models are the sole judge (Kalai & Vempala, 2024). A related signal is the persistent gap between human annotators and LLMs on hallucinated-text classification benchmarks (Lin et al., 2022; Li et al., 2023).

Image-generation studies also routinely report quantitative perceptual metrics such as FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021), many of which summarize feature embeddings from pretrained networks such as Inception-v3 (Szegedy et al., 2016). These scores are valuable for distributional or instance-level visual quality, but they do not specify whether a generated image preserves a particular factual count. We therefore treat *whether these metrics capture hallucination severity* as an empirical question and answer it through correlation analyses against CHR and NCFR.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $\mathbf{x}_0 \sim q_0(\mathbf{x}_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define a *forward process* (a.k.a. *diffusion process*) that adds Gaussian noise across T discrete timesteps. This process fixes the noise scale used later in our solver comparisons: at timestep t , the marginal distribution of the noisy sample $\mathbf{x}_t$ conditioned on $\mathbf{x}_0$ is:

$$q_{0:t}(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t \mid \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

Here $t \in \{0, 1, \dots, T\}$, and $\{\beta_t\}_{t=1}^T$ is a pre-defined timestep-dependent noise schedule, commonly linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021). The coefficient $\bar{\alpha}_t$ determines the signal-to-noise ratio at each step, moving from $\bar{\alpha}_0 = 1$ (no noise) to $\bar{\alpha}_T \approx 0$ (nearly pure noise). Equivalently, Eq. (1) can be written as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + (1 - \bar{\alpha}_t)\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

For sufficiently large T , the terminal marginal $q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ approaches the standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$ used as the sampling prior. Widely used DPMs (Ho et al., 2020; Nichol & Dhariwal, 2021) typically set T to 1000, which is also the ancestral-sampling step count used in our DDPM comparisons.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (a.k.a., *denoising process*) recovers data by iteratively mapping the diffused variable $\mathbf{x}_T$ back toward $\mathbf{x}_0$. DPMs implement this step by training a neural network $\epsilon_\theta(\mathbf{x}_t, t)$, the *noise prediction model*, to estimate the noise ϵ from $\mathbf{x}_t$. The parameters θ are optimized with the standard objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\pi(t) \|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a weighting function.

Ancestral Sampling. Given noisy data $\mathbf{x}_T$, ancestral sampling estimates samples from $q_0(\mathbf{x}_0)$ by reversing the diffusion chain with the same step count as the forward process:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1-\beta_t}}(\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

The only unknown quantity is $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$, the *score function* (Song et al., 2022).

In practice, DPMs use the noise prediction model to approximate the scaled score $-\sqrt{(1-\bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$. Using the objective in Eq. (3), this gives the equivalent update form used to compare ancestral sampling against accelerated ODE solvers:

$$\mathbf{x}_{t-1} = \mu_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon, \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. Ancestral sampling in Eq. (4) contains a stochastic term (Kloeden et al., 1992). For accelerated inference, the associated deterministic process can be written as an ordinary differential equation (ODE), also called the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau)\mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2}\beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2}\beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ be the ideal trajectory satisfying this ODE with the true score function $h(\mathbf{x}(\tau), \tau)$ and an ideal diffused initialization $\mathbf{x}(T)$ drawn from $q_T(\mathbf{x}(T))$. We set discrete timesteps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, with $\tau_N = 0$. The exact one-step solution follows from the variation-of-constants formula:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

Here $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear component $\frac{dz}{d\tau} = f(\tau)z$. The step size $\Delta t > 0$ determines the number of sampling steps $T/\Delta t$, so larger Δt enables faster ODE-based sampling than ancestral sampling. Because $\mathcal{G}(t_2, t_1)$ is available in closed form, accelerated samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) mainly differ in how accurately they approximate the nonlinear integral involving h , through choices such as solver order and step count. These are precisely the numerical factors evaluated in our CHR, NCFR, and TFR comparisons; additional analysis appears in Appendix A.1 to A.3.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section specifies the measurement design used to turn counting hallucinations into comparable rates. We first define dataset-level counting criteria and the deviations that constitute counting hallucinations, then introduce the CountHalluSet datasets. For each dataset, we state the valid count rule, the model used to estimate counts, and the evaluation condition under which CHR and related failure rates are computed.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

We define counting hallucinations relative to explicit counting criteria in the reference dataset, so every generated sample is evaluated against the same measurable constraint.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ denote the reference training dataset. For each sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$ and category c , the valid object count $N_c(\mathbf{x})$ must lie in a pre-defined set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$), so $N_c(\mathbf{x}) \in \mathcal{S}_c$. We additionally require each valid reference image to contain at least one object, $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C is the set of evaluated categories.

Counting hallucinations as deviations from the counting criteria. For a generated sample $\hat{\mathbf{x}}$, we count a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ only when the sample satisfies both requirements below:

1. **Sufficient visual quality for counting.** The sample $\hat{\mathbf{x}}$ must have enough visual quality for the relevant objects and background to be discernible. This condition separates count-specific factual errors from broad non-counting failures such as severe degradation. We encode this decision with $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, the *counting-ready indicator*, where 1 means the sample is suitable for counting and 0 means it is excluded from count-error attribution. In practice, the indicator may come from human annotation, a classifier calibrated to human judgments, or another automated rule.
2. **Violation of counting facts.** The predicted counts must violate the reference rule: either some category c has $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or the image contains no objects ($\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$) despite the dataset requiring at least one object. We use the dataset-specific counting model to provide the empirical count evidence for this comparison.

Thus, CH is assigned only when the sample is counting-ready ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and the predicted count falls outside the dataset rule ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

CountHalluSet contains three datasets that make the comparison increasingly realistic: ToyShape, SimObject, and RealHand. Examples of reference samples and matching hallucination types are shown in Fig. A1 and A2, while Appendix A.4 reports category and object-count statistics used to verify dataset balance.

4.2.1 TOYSHAPE

Dataset description. ToyShape provides the simplest quantitative setting: 30,000 images containing three geometric categories, *triangle*, *square*, and *pentagon*. All shapes are white, have equal area of 120 pixels, and never overlap, minimizing non-counting ambiguity. The counting rule is explicit: each category may appear at most once, and each image must contain at least one shape, as shown in the top-left of Fig. 1. Formally, $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$ and, for each class $c \in C$, $N_c(\mathbf{x}) \in \mathcal{S}_c = \{0, 1\}$ subject to $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. Under this rule, a ToyShape generation is counted as hallucinated if it contains two or more instances of any category, such as the two-pentagon example in the top-right of Fig. 1. An empty image with only the black background also violates the at-least-one-object requirement and is therefore counted as a hallucination. Representative examples appear in Appendix A.5.

Counting model used in ToyShape. ToyShape does not require a counting-ready indicator because severe visual failures are rare in this controlled setting, so every generated image is evaluated directly by a counting model. We fine-tune ResNet-50 (He et al., 2016) on over 400k ToyShape samples in which each category appears 0–3 times per image (i.e., $\mathcal{S}_c = \{0, 1, 2, 3\}$), covering both valid counts and hallucinated counts. This counting model achieves over 99.9% accuracy on generated data, supporting reliable CHR estimation.

Quantification of counting hallucinations. For each diffusion model trained on ToyShape, we generate 30,000 samples, matching the training-set size. The counting model predicts per-category object counts for every generated image, and CHR is computed from the fraction that violates the ToyShape rule, as illustrated in Fig. 2.

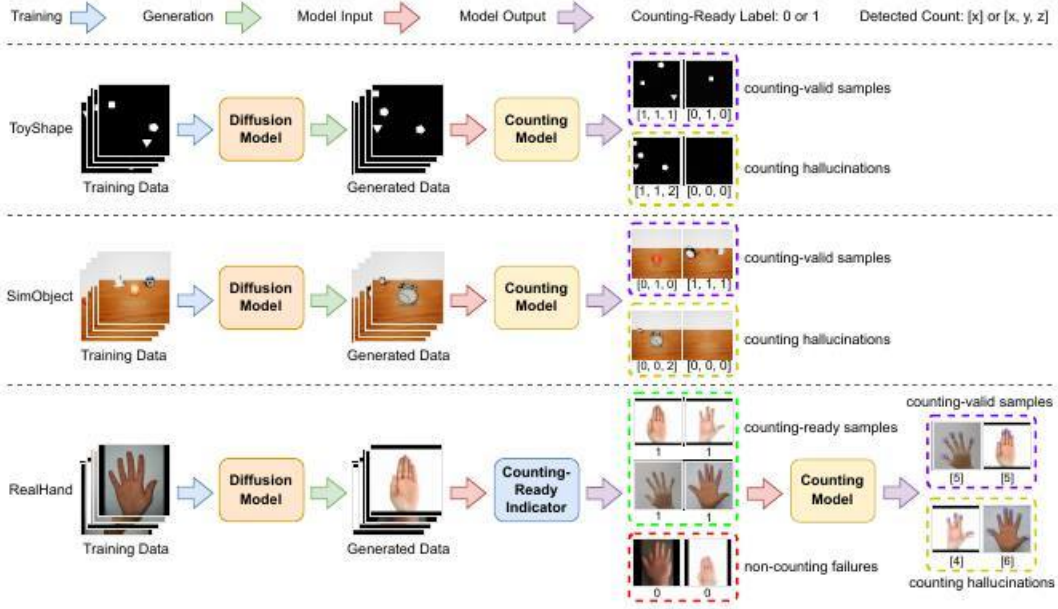


Figure 2: CountHalluSet evaluation pipeline. ToyShape and SimObject are simple enough that all generated images are directly counted. RealHand adds a counting-ready indicator, calibrated to human judgments, to separate count-specific hallucinations from non-counting failures such as severely deformed fingers. Count labels are ToyShape: *triangle*, *square*, *pentagon*; SimObject: *mug*, *apple*, *clock*; and RealHand: *finger*.

4.2.2 SIMOBJECT

Dataset description. SimObject preserves the same count-rule structure as ToyShape while increasing visual realism: it contains 30,000 rendered images of everyday objects. The categories are *mug*, *apple*, and *clock*, each with 10 intra-class variations, placed randomly on a wooden tabletop under fixed lighting and with minimal mutual occlusion, as shown in the middle-left of Fig. 1. The images are generated in Unreal Engine 5 (Epic Games, 2024), providing photorealistic geometry, materials, lighting, and reflections. The counting criteria match ToyShape (Sec. 4.2.1): $C = \{mug, apple, clock\}$ and, for each $c \in C$, $N_c(\mathbf{x}) \in S_c = \{0, 1\}$ subject to $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. Because SimObject uses the same valid-count rule as ToyShape, we apply the same hallucination definition, counting-model construction, and direct quantification protocol, as summarized in Fig. 2.

4.2.3 REALHAND

Dataset description. RealHand tests whether the same quantitative framing holds for real-world structure by using 5,050 human hand images collected from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². We aggregate raw images from these sources and remove images with blur or artifacts. Each retained image shows a five-finger human hand, including dorsal (back) and palmar (front) views across varied lighting and backgrounds such as white surfaces, blackboards, and walls, as shown in the lower-left of Fig. 1. The count rule is stricter than in the synthetic datasets: $C = \{finger\}$ with $N_c(\mathbf{x}) \in S_c = 5$ for each $c \in C$, and $\sum_{c \in C} N_c(\mathbf{x}) = 5$.

Counting hallucinations and non-counting failures. RealHand also makes it necessary to distinguish count-specific errors from broader image failures. Generated hands can contain extra or missing fingers, but they can also be blurry, severely deformed, or artifacted in ways that make counting ill-posed; we therefore use a counting-ready indicator, implemented as a binary classifier, to separate these outcomes. Examples of both counting hallucinations and non-counting failures appear in Appendix A.7.

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

Counting-ready indicator and counting model used in RealHand. To operationalize this separation, we finetune MaxViT (Tu et al., 2022) on 2.5k annotated generated images as the counting-ready indicator aligned with human judgments. For fingertip counting, we finetune YOLO-12 (Tian et al., 2025) on 2k generated images annotated with fingertip locations. On a human-evaluated set of 500 randomly sampled images labeled by three annotators, the indicator and counting model achieve over 96% and 99% accuracy, respectively, providing reliable measurement for CHR and NCFR. YOLO detection examples are shown in Appendix A.6.

Quantification of counting hallucinations. ToyShape and SimObject can be evaluated directly because severe non-counting failures are rare. RealHand requires one additional step: we first select counting-ready hand images with the indicator, then apply the counting model to estimate finger count and compute CHR among samples suitable for count attribution, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. For the model comparisons, we train DDPM (Ho et al., 2020) from scratch on ToyShape and fine-tune latent diffusion models (LDMs) (Rombach et al., 2022), pretrained on CelebA-HQ (Karras et al., 2018), on SimObject and RealHand. All diffusion models use $T = 1,000$ with a standard linear noise schedule, making ancestral DDPM sampling directly comparable to accelerated ODE sampling. Images are resized to 128×128 for DDPMs and 256×256 for LDMs. Using Adam (Kingma, 2015) with learning rate 0.0001 and effective batch size 256, we train for 150k steps on ToyShape, 300k on SimObject, and 80k on RealHand.

Inference of Diffusion Models. At inference, DDPM uses 1,000 ancestral sampling steps. For accelerated ODE sampling, we compare a first-order solver, DPM-Solver-1 (equivalent to DDIM (Song et al., 2021)), with a second-order solver, DPM-Solver-2 (Lu et al., 2022). Each ODE solver is evaluated at 25, 50, and 100 denoising steps, so trends can be read as step-controlled and order-controlled comparisons. For RealHand, YOLO counting uses confidence threshold 0.3 and IoU threshold 0.1. Unless otherwise stated, each evaluation generates the same number of samples as the corresponding training set, and reported results average over three random seeds.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports the central quantitative comparison: CHR for ToyShape, SimObject, and RealHand, and additionally NCFR and TFR for RealHand. Rows vary solver type and sampling steps, while paired columns compare diffused versus normal initial noise. This layout makes the empirical patterns visible across dataset complexity, ODE solver order, denoising budget, and initialization.

(1) Increasing sampling steps reduces counting hallucinations in synthetic datasets but exacerbates them in RealHand. In Table 1, increasing ODE sampling steps from 25 to 50 to 100 generally lowers CHR for ToyShape and SimObject, showing the expected benefit of finer denoising on simpler synthetic distributions. The RealHand comparison moves differently: CHR typically rises as steps increase, with the exception of DPM-Solver-2 at 100 steps. Thus, the practical effect of additional steps depends on dataset structure; for morphologically complex hands, more steps can improve aspects of generation while failing to improve, and sometimes worsening, finger-count factuality.

(2) Employing higher-order ODE solvers effectively mitigates non-counting failures but exacerbates counting hallucinations in RealHand under the standard inference steps. The RealHand columns in Table 1 show a clear trade-off between DPM-Solver-1 and DPM-Solver-2 at standard step counts such as 25 and 50. The higher-order solver reduces NCFR, indicating fewer visually unsuitable samples, but it also yields higher CHR, indicating more count-specific finger errors among counting-ready outputs. The same distinction appears when increasing steps within a solver: more computation can reduce non-counting failures without enforcing the exact five-finger rule.

(3) DDPM substantially outperforms ODE solvers in mitigating both counting hallucinations and non-counting failures. Across the reported comparisons, ancestral DDPM sampling achieves the lowest CHR, NCFR, and TFR. For example, RealHand with normal noise drops to 10.75 CHR and

Table 1: Quantitative comparison of failure rates under matched solver, step, dataset, and initialization conditions. CHR is the number of counting hallucinations divided by generated samples for ToyShape, SimObject, and RealHand. For RealHand, NCFR reports samples rejected by the counting-ready indicator, and TFR is CHR+NCFR. *Diffused* and *Normal* compare ground-truth initial noise against standard Gaussian noise.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

2.39 NCFR under DDPM, below the corresponding ODE-solver entries in the same table. This makes DDPM a strong empirical reference point for the lowest observed failure rates in our evaluation, albeit at the cost of 1,000 sampling steps.

(4) **A better initial noise configuration can reduce both counting hallucinations and non-counting failures.** Across most paired entries in Table 1, “diffused” noise (ground-truth initial noise) yields lower CHR, NCFR, and TFR than “normal” noise (standard Gaussian noise). The paired-column comparison suggests that initialization quality affects both count-specific factuality and broader visual failures, likely because diffused noise is better aligned with the training-time initial noise distribution.

(5) **Greater morphological complexity in countable objects increases the likelihood of counting hallucinations.** The dataset-wise CHR comparison increases with morphological complexity: simple geometric shapes have the lowest rates, photorealistic tabletop objects are higher, and hands are highest among the three settings. This pattern highlights practical significance beyond individual numbers: maintaining correct counts becomes harder when the countable unit is defined by fine-grained parts and spatial relationships rather than by isolated simple objects.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Counting hallucination is a factual error, so the key metric question is quantitative rather than semantic: *does a better perceptual metric, such as lower FID, correspond to a lower CHR under the same dataset and solver conditions?* The empirical answer is **not necessarily**.

We therefore correlate CHR and NCFR separately with the same perceptual metrics. The metrics are FID (Heusel et al., 2017), a distribution-level measure, and MUSIQ (Ke et al., 2021), an instance-level quality measure. For each relationship, we report Pearson correlation (Lee Rodgers & Nicewander, 1988) for linear dependence, Spearman rank correlation (Spearman, 1904) for monotonic dependence, and Distance Correlation (Székely et al., 2007) for general nonlinear dependence. This comparison reveals two distinct empirical regimes.

✚ **The correlations between CHR and perceptual metrics are dataset-dependent and solver-dependent rather than intrinsic.** In Table 2, CHR–FID correlations change sign and magnitude across evaluation conditions: Pearson correlation is positive for SimObject, negative for RealHand, and weakly positive for PTI-Hand, while Spearman correlation shifts from positive to negative to negative. The distance-correlation row with integrated DDPM and ODE results also falls from 0.82 on SimObject to 0.74 on RealHand, and the PTI-Hand distance correlation is 0.47. These margins show that CHR is not an intrinsic monotone function of FID; its relationship to perceptual quality depends on dataset and solver regime.

✚ **The correlations between NCFR and perceptual metrics are consistently strong and positive.** By contrast, NCFR aligns strongly and consistently with perceptual metrics. For NCFR versus FID, the reported coefficients are all above 0.88 with very small p-values, indicating that FID captures broad visual failure much more reliably than it captures count-specific factual errors.

Table 2: Correlation analysis separating count-specific failures from general visual failures. The table reports CHR, NCFR, and TFR relationships with FID and MUSIQ across datasets and sampling conditions. PTI-Hand uses ten prompts with Stable-Diffusion-3.5-Medium to generate 5050 samples, matching RealHand size; PTI-Hand FID is computed against RealHand. Here, p is the significance value for r or ρ , and $\dagger$ indicates correlations after integrating DDPM results with ODE-solver results across sampling conditions.

Metric	CHR vs. FID				NCFR vs. FID				CHR vs. MUSIQ				NCFR vs. MUSIQ			
	SimObject	RealHand	PTI-Hand		RealHand	PTI-Hand			RealHand	PTI-Hand			RealHand	PTI-Hand		
	r	ρ	r	ρ	r	ρ	r	ρ	r	ρ	r	ρ	r	ρ	r	ρ
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01
Pearson $\dagger$ (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01
Spearman $\dagger$ (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.93	—	0.42	—	0.97	—	0.99	—
Distance Correlation $\dagger$ (r_d)	0.82	—	0.74	—	—	—	0.95	—	0.71	—	—	—	0.97	—	—	—

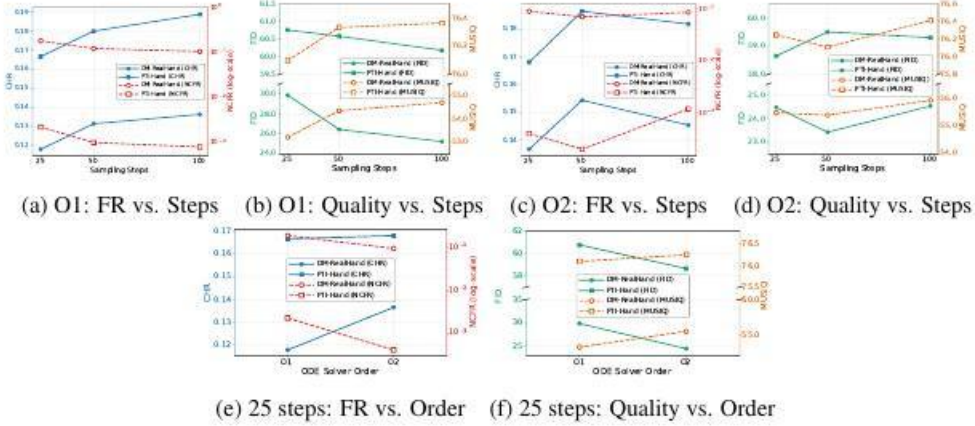


Figure 3: Trend comparison between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium). O1 and O2 denote first- and second-order ODE solvers (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). FR metrics are CHR and NCFR, while quality metrics are FID and MUSIQ. The paired plots show that step- and order-dependent trends are consistent across the controlled RealHand model and the pretrained text-to-image model.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

We next test whether the RealHand patterns transfer to a large-scale pretrained text-to-image model. Stable-Diffusion-3.5-Medium (Esser et al., 2024) is evaluated with the same counting-ready indicator and counting model introduced in Sec. 4.2.3; Appendix A.8 gives the prompt details. As shown in Figure 3, changes in solver type and sampling steps produce trends that closely match the controlled DM-RealHand experiments in Sec. 5.2. The correlation evidence in Table 2 reinforces the same pattern: CHR remains weakly tied to perceptual metrics, while NCFR remains strongly tied to them. Thus, the empirical comparison is not confined to a small trained model; the same measurement protocol and qualitative trend structure transfer to pretrained text-to-image generation.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

Observation 5 in Sec. 5.2 shows lower CHR for simpler countable structures, suggesting that diffusion models preserve count facts more reliably when the relevant topology is easier to model. This empirical pattern motivates a practical intervention: denoise RGB appearance jointly with an explicit structural representation, such as an object mask that encodes basic topology and counts, in the same latent space. We call models trained with this joint visual-structural denoising objective **joint-diffusion models** (JDM).

Concretely, JDM concatenates the latent representation of each hand mask with the latent representation of the corresponding RGB image along the channel dimension, as shown in Fig. 4. We then fine-tune the CelebA-HQ-pretrained UNet (Karras et al., 2018), using the same RealHand

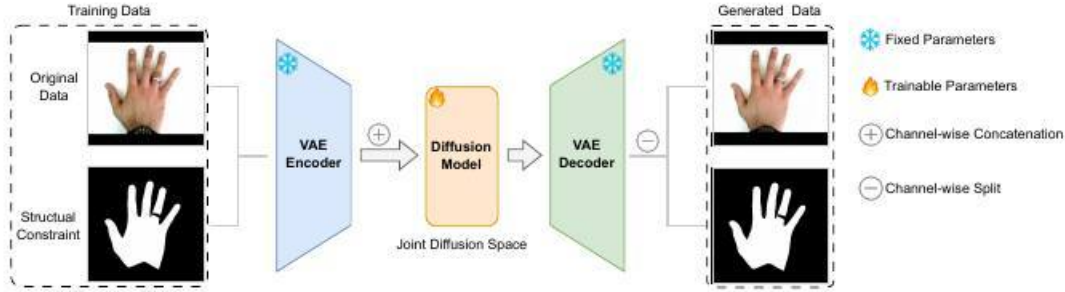


Figure 4: Joint-diffusion model (JDM) architecture used for the RealHand comparison. JDM concatenates hand RGB latents and segmentation-mask latents channel-wise, so denoising is conditioned on both appearance and explicit structural topology. The resulting RealHand evaluations show lower CHR and NCFR than vanilla LDM across the reported solver and step settings.

Table 3: RealHand comparison between vanilla LDM and JDM across ODE solver order and sampling steps. Lower CHR, NCFR, and FID are better. JDM reduces CHR and NCFR in every reported solver-step condition, while FID is computed only on RGB outputs, despite joint RGB-mask generation, to keep the perceptual metric comparable with the LDM baseline.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

configuration as Sec. 5.1, on RealHand images and SAM-2-extracted masks (Ravi et al., 2024). The denoising UNet input and output channels are expanded from the RGB latent dimensionality to the combined image-mask dimensionality (i.e., 3+3=6 channels).

Table 3 shows the practical margin of this intervention: across all listed solvers and step counts, JDM has lower CHR and lower NCFR than the vanilla LDM on RealHand. For example, under DPM-Solver-2 at 50 steps, CHR decreases from 15.99 to 9.66 and NCFR from 7.22 to 4.63, while other columns show the same direction of improvement. The trade-off is higher FID, likely because joint RGB-mask denoising increases the modeling burden in a small-scale training regime. Thus, JDM improves count and failure-rate outcomes while sacrificing some perceptual-distribution score, consistent with our finding that FID and CHR need not move together. Additional details and pretrained-model finetuning results appear in Appendix A.9.

6 CONCLUSION

This work reframes diffusion-model hallucination as a measurable empirical quantity by focusing on counting hallucinations and reporting them under controlled comparisons. We introduce **CountHalluSet** and a reproducible protocol that specifies the dataset rule, counting model, baseline, solver, initialization, and metric for each evaluation. The resulting evidence shows that standard numerical improvements, such as more ODE steps or higher-order solvers, can reduce non-counting failures while leaving RealHand counting hallucinations unchanged or worse. Correlation analyses further show that FID and MUSIQ track NCFR much more consistently than CHR, so perceptual metrics should not be treated as substitutes for count-based factual evaluation. Finally, the PTI-Hand and JDM experiments demonstrate practical significance: the observed trends transfer to pretrained text-to-image models, and joint RGB-mask denoising reduces both CHR and NCFR on RealHand despite a higher FID trade-off.

REFERENCES

Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J Ballard, Joshua Bambrick, et al. Accurate structure prediction of biomolecular interactions with alphafold 3. *Nature*, pp. 493—500, 2024.

- Mahmoud Afifi. 11k hands: Gender recognition and biometric identification using a large dataset of hand images. *Multimedia Tools and Applications*, 78:20835–20854, 2019.
- Sumukh K Aithal, Pratyush Maini, Zachary C Lipton, and J Zico Kolter. Understanding hallucinations in diffusion models through mode interpolation. *arXiv preprint arXiv:2406.09358*, 2024.
- Ali Borji. Qualitative failures of image generation models and their application in detecting deepfakes. *Image and Vision Computing*, 137:104771, 2023.
- Jake Bruce, Michael D Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, et al. Genie: Generative interactive environments. In *ICML*, 2024.
- Thomas M Cover and Joy A Thomas. *Elements of Information Theory*. John Wiley & Sons, 2012.
- Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *NeurIPS*, 34:8780–8794, 2021.
- Jingtao Ding, Yunke Zhang, Yu Shang, Yuheng Zhang, Zefang Zong, Jie Feng, Yuan Yuan, Hongyuan Su, Nian Li, Nicholas Sukiennik, et al. Understanding world or predicting future? a comprehensive survey of world models. *ACM Computing Surveys*, 2024.
- Epic Games. Unreal engine, 2024. URL <https://www.unrealengine.com>.
- Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *ICML*, pp. 12606–12633, 2024.
- David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2(3), 2018.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pp. 770–778, 2016.
- Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. In *EMNLP*, pp. 7514–7528, 2021.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *NeurIPS*, 30, 2017.
- Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*, pp. 1–14, 2021.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *NeurIPS*, 33: 6840–6851, 2020.
- Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J Fleet. Video diffusion models. *NeurIPS*, 35:8633–8646, 2022.
- Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366, 1989.
- Kaiyi Huang, Chengqi Duan, Kaiyue Sun, Enze Xie, Zhenguo Li, and Xihui Liu. T2i-compbench++: An enhanced and comprehensive benchmark for compositional text-to-image generation. *IEEE TPAMI*, 2025.
- Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232*, 2023.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM computing surveys*, 55(12):1–38, 2023.

- John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Židek, Anna Potapenko, et al. Highly accurate protein structure prediction with alphafold. *Nature*, 596(7873):583–589, 2021.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Adam Tauman Kalai and Santosh S Vempala. Calibrated language models must hallucinate. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, pp. 160–171, 2024.
- Tero Karras, Timo Aila, Samuli Laine, and Jaakko Lehtinen. Progressive growing of gans for improved quality, stability, and variation. In *ICLR*, 2018.
- Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. Musiq: Multi-scale image quality transformer. In *ICCV*, pp. 5148–5157, 2021.
- Diederik P Kingma. Adam: A method for stochastic optimization. In *ICLR*, pp. 1–15, 2015.
- Peter E Kloeden, Eckhard Platen, Peter E Kloeden, and Eckhard Platen. *Stochastic differential equations*. Springer, 1992.
- Zhifeng Kong, Wei Ping, Jiaji Huang, Kexin Zhao, and Bryan Catanzaro. Diffwave: A versatile diffusion model for audio synthesis. In *ICLR*, pp. 1–17, 2021.
- Joseph Lee Rodgers and W Alan Nicewander. Thirteen ways to look at the correlation coefficient. *The American Statistician*, 42(1):59–66, 1988.
- Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. Halueval: A large-scale hallucination evaluation benchmark for large language models. In *EMNLP*, pp. 6449–6464, 2023.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *ACL*, pp. 3214–3252, 2022.
- Luping Liu, Yi Ren, Zhijie Lin, and Zhou Zhao. Pseudo numerical methods for diffusion models on manifolds. In *ICLR*, pp. 1–24, 2022.
- Qihao Liu, Adam Kortylewski, Yutong Bai, Song Bai, and Alan Yuille. Discovering failure modes of text-guided diffusion models via adversarial search. In *ICLR*, pp. 1–25, 2024.
- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver: A fast ode solver for diffusion probabilistic model sampling in around 10 steps. *NeurIPS*, 35:5775–5787, 2022.
- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver++: Fast solver for guided sampling of diffusion probabilistic models. In *ICLR*, pp. 1–19, 2023.
- Shitong Luo and Wei Hu. Diffusion probabilistic models for 3d point cloud generation. In *CVPR*, pp. 2837–2845, 2021.
- Supreeth Narasimhaswamy, Uttaran Bhattacharya, Xiang Chen, Ishita Dasgupta, Saayan Mitra, and Minh Hoai. Handdiffuser: Text-to-image generation with realistic hand appearances. In *CVPR*, pp. 2468–2479, 2024.
- Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *ICML*, pp. 8162–8171, 2021.
- Michael A Nielsen. *Neural networks and deep learning*, volume 25. Determination press San Francisco, CA, USA, 2015.
- Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.

-
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, pp. 10684–10695, 2022.
- Mehdi SM Sajjadi, Olivier Bachem, Mario Lucic, Olivier Bousquet, and Sylvain Gelly. Assessing generative models via precision and recall. *NeurIPS*, 31, 2018.
- Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training gans. *NeurIPS*, 29, 2016.
- Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *ICML*, pp. 2256–2265, 2015.
- Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *ICLR*, pp. 1–22, 2021.
- Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *ICLR*, pp. 1–36, 2022.
- C Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904.
- Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *CVPR*, pp. 2818–2826, 2016.
- Gábor J Székely, Maria L Rizzo, and Nail K Bakirov. Measuring and testing dependence by correlation of distances. *The Annals of Statistics*, pp. 2769–2794, 2007.
- Yunjie Tian, Qixiang Ye, and David Doermann. Yolov12: Attention-centric real-time object detectors. *arXiv preprint arXiv:2502.12524*, 2025.
- Zhengzhong Tu, Hossein Talebi, Han Zhang, Feng Yang, Peyman Milanfar, Alan Bovik, and Yinxiao Li. Maxvit: Multi-axis vision transformer. In *ECCV*, pp. 459–479, 2022.
- Arash Vahdat, Francis Williams, Zan Gojcic, Or Litany, Sanja Fidler, Karsten Kreis, et al. Lion: Latent point diffusion models for 3d shape generation. *NeurIPS*, 35:10021–10039, 2022.
- Shirley Wu, Mert Yuksekgonul, Linjun Zhang, and James Zou. Discover and cure: Concept-aware mitigation of spurious correlation. In *ICML*, pp. 37765–37786, 2023.
- Dmitry Yarotsky. Universal approximations of invariant maps by neural networks. *Constructive Approximation*, 55(1):407–474, 2022.
- Arman Zarei, Keivan Rezaei, Samyadeep Basu, Mehrdad Saberi, Mazda Moayeri, Priyatham Kattakinda, and Soheil Feizi. Understanding and mitigating compositional issues in text-to-image generative models. *arXiv preprint arXiv:2406.07844*, 2024.
- Qinsheng Zhang and Yongxin Chen. Fast sampling of diffusion models with exponential integrator. In *ICLR*, pp. 1–33, 2023.

A APPENDIX

A.1 POOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the appendix error analysis, we compare the ideal reverse process $q(\mathbf{x}_{T:0})$ with the model reverse process $p_\theta(\mathbf{x}_{T:0})$ using KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

Using the Markov property, both trajectory distributions decompose into an initial terminal distribution and per-step reverse transitions:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

Applying the chain rule makes the two error sources explicit in the KL comparison:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \middle| \mathbf{x}_t \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

The Data Processing Inequality (Cover & Thomas, 2012) then links trajectory-level discrepancy to sample-level discrepancy: marginalizing variables cannot increase KL divergence.

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

Combining Eq. (A.4) and Eq. (A.5) yields the bound on total accumulated distributional error:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

Equality would hold only in the ideal comparison where $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for every t and $\mathbf{x}_t$, meaning the model exactly recovers $q(\mathbf{x}_0)$. When data entropy $H(\mathbf{x}_0)$ is excluded, this upper bound on $\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0))$ matches the DDPM variational bound (Ho et al., 2020), so optimizing the DDPM objective tightens an explicit discrepancy bound between true and generated data.

Diffused-prior gap. The first error term is the mismatch between the ground-truth diffused distribution $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ and the sampling prior $p_\theta(\mathbf{x}_T)$. Because practical models use finite T , this diffused-prior gap is not exactly zero. A distributional metric can quantify the mismatch:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

Here $\mathcal{D}_{\text{KL}}$ is the Kullback–Leibler (KL) divergence. Other distributional metrics, including FID or Wasserstein distance, could also be used to summarize this initial discrepancy. Because the gap is present before denoising begins, it can propagate through the reverse process and contribute to final output error.

Noise model prediction error. The second error term comes from the noise prediction model. The estimate $\epsilon_\theta(\mathbf{x}_t, t)$ approximates, but need not equal, the true noise ϵ , even under the expressive capacity suggested by the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Let $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ denote this prediction error. From Eq. (5), any error in ϵ_θ produces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Because each reverse transition depends on the previous state, these mean deviations accumulate along the denoising trajectory and contribute to the final generation error.

A.2 POOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Following Eq. (7), we illustrate accumulated ODE error with a one-step numerical solver. Let $\mathbf{x}^*(\tau)$ be the ideal trajectory using the true score function $h^*(\mathbf{x}(\tau), \tau)$ and the ground-truth initialization $\mathbf{x}^*(T) \sim q_T(\mathbf{x}_T)$. Let $\hat{\mathbf{x}}_k$ be the numerical trajectory at timesteps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, and let $\hat{h}$ be the model-estimated function. For a trajectory from $\hat{\mathbf{x}}_0$ to $\hat{\mathbf{x}}_N$, the accumulated trajectory error is tracked by $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ over transitions from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. The exact ideal one-step evolution from τ_k down to τ_{k+1} follows the variation-of-constants formula:

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

Here $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear component $\frac{d\mathbf{z}}{d\tau} = f(\tau)\mathbf{z}$, and the integration limits proceed backward in time.

Numerical path evolution. A one-step numerical solver approximates, rather than exactly follows, the evolution induced by $\hat{h}$ from $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h})$ denote the exact solution at τ_{k+1} for the ODE using $\hat{h}$ and starting from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The difference between the numerical update and this exact $\hat{h}$ -based evolution is the local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

Therefore the numerical update can be written as:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtracting the ideal evolution in Eq. (A.9) from the numerical evolution in Eq. (A.12) gives the recurrence:

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. Unrolling this recurrence from $k = 0$ to $N - 1$ gives the accumulated trajectory error. Let $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, with $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$. Repeated application yields:

$$\begin{aligned} e_N &= \mathcal{G}(\tau_N, \tau_0) e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \\ &\quad \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \end{aligned} \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$ makes the final-sample error explicit:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

The propagated initial-error term is the deterministic contribution of a particular initialization mismatch to the final path error, with its statistical source in the diffused-prior gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

For comparing DDPM and DDIM, the DDPM (Ho et al., 2020) forward process from Eq. (2) is:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

The DDPM reverse update is:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon. \quad (\text{A.17})$$

DDIM (Song et al., 2021) uses the same forward process as Eq. (A.16), but its deterministic reverse process with $\eta = 0$ uses a different update:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\alpha_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \epsilon_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Thus, even though DDPM and DDIM share the forward diffusion process, DDIM is not merely DDPM with the stochastic term removed. Their reverse updates weight the predicted noise term with different coefficients, which is relevant when comparing solver behavior empirically.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 shows representative reference images for the three datasets used in the quantitative evaluation. Table A1 reports category-level counts, and Table A2 reports image counts by object-count condition; together, these summaries show that the synthetic datasets are balanced across categories and object-count levels, while RealHand consistently contributes five fingers per image. The datasets use the same order in both tables, and the category order in Table A1 is:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

Fig. A2 provides representative hallucination cases for all three datasets, illustrating the exact visual evidence counted by the protocol. For ToyShape and SimObject, hallucinations either duplicate a category beyond the allowed single instance or contain no object despite a normal background, as shown in Fig. A2(a) and Fig. A2(b). For RealHand, hallucinations are samples whose valid finger count differs from five, such as 4 or 6 fingers, as shown in Fig. A2(c). The last two images of the first row in Fig. A2(c) visually resemble five-finger hands, but one finger is incomplete and artifacted, so the valid count is four and the samples are classified as counting hallucinations.

Table A1: Category-level object counts used to check dataset balance before evaluating counting hallucination rates.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Image counts by valid object-count condition for each dataset, showing the reference distribution used for CHR comparisons.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 shows representative YOLO fingertip predictions used in the RealHand evaluation. The detector identifies fingertips across varied poses and orientations, including dorsal and palmar views, which supports the reported counting accuracy. A remaining limitation is anatomical rather than numeric: hands with duplicated thumbs, such as examples in Fig. A3(a), or five middle fingers and no thumb, as also illustrated in Fig. A3(a), can still be detected as five-finger samples. Such samples are count-valid under the current metric because they match the five-fingertip count, even if they are anatomically implausible.

Figure A3(b) shows counting-hallucinated samples where YOLO predicts 3, 4, 6, or 7 fingertips, providing the direct count evidence used for CHR. Figure A3(c) shows rare edge cases (<1%) involving unclear boundaries, cracked fingers, or floating artifacts where detection can fail. Adjusting IoU or confidence thresholds could reduce some of these cases, but may hurt performance on the dominant counting-valid population. Because the same detector configuration is used for every RealHand comparison, these rare cases do not change the relative conclusions reported in the main experiments.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 illustrates the three RealHand outcome categories used to compute NCFR and CHR: counting-valid, counting-hallucinated, and non-counting failure. As summarized in Fig. 2, the counting-ready indicator first separates samples suitable for count attribution from images with severe artifacts, such as distorted, cracked, partially visible, or floating fingers. Only counting-ready samples are then classified by the fingertip count, with exactly five valid fingertips counted as valid and any other valid count counted as hallucinated.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

For PTI-Hand, we use 10 hand-generation prompts with Stable-Diffusion-3.5-Medium so the pretrained-model comparison matches the RealHand evaluation size. The prompts are:

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person's hand, entire hand in frame, all fingers visible, plain background.

5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

Counting hallucination is defined relative to patterns absent from training data, but the full training distribution of a large pretrained model is inaccessible. For PTI-Hand, we therefore use a conservative count rule: any generated hand with six or more fingers is counted as a hallucination. This rule avoids over-claiming on images with fewer visible fingers, which can occur naturally through occlusion, articulation, or viewpoint, while still capturing an empirically reliable signal because real hands with six or more fingers are highly unlikely in ordinary training data. Generated PTI-Hand examples are shown in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

For the JDM results in Table 3, the denoising UNet is fine-tuned for 120k steps, compared with 80k steps for vanilla LDM. The longer training budget reflects the harder optimization problem introduced by expanding the latent space from three to six channels, which creates a mismatch with the pretrained LDM latent distribution. Additional optimization is therefore used to reduce this distributional gap before comparing CHR, NCFR, and FID.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

Table A3: RealHand comparison between finetuned SD-1.5 and SD-1.5 with joint-diffusion finetuning. The baseline row fine-tunes Stable-Diffusion-1.5 on RealHand. The joint-diffusion row freezes intermediate layers and fine-tunes only the extended-channel input and output layers. Lower CHR, NCFR, and FID indicate better performance, enabling direct comparison of count-specific, non-counting, and perceptual outcomes.

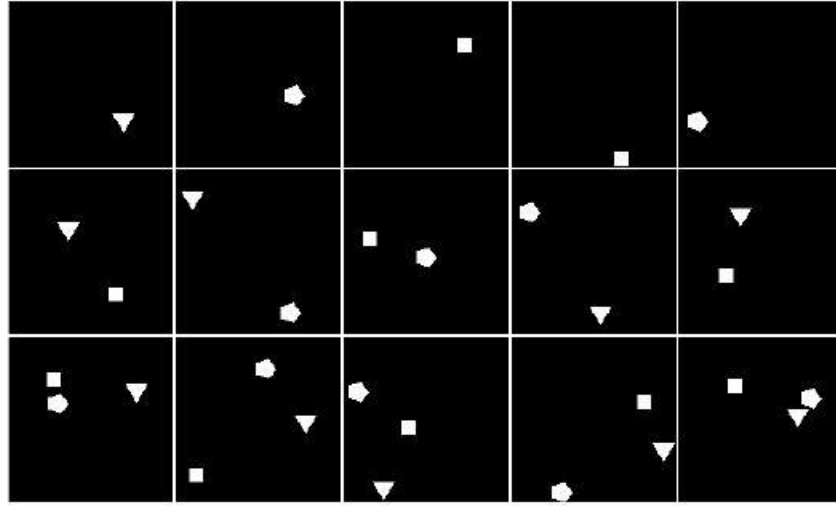
Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.1404	0.3630	0.3796	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

JDM should therefore be interpreted as a **training paradigm** for improving factual generation, not as a universally superior inference strategy. Its inference-time overhead is small because it extends only the UNet input and output channels while leaving intermediate layer sizes unchanged. However, adding mask channels to a pretrained text-to-image model disrupts the geometry of the pretrained latent manifold and the original denoising distribution, making direct finetuning nontrivial.

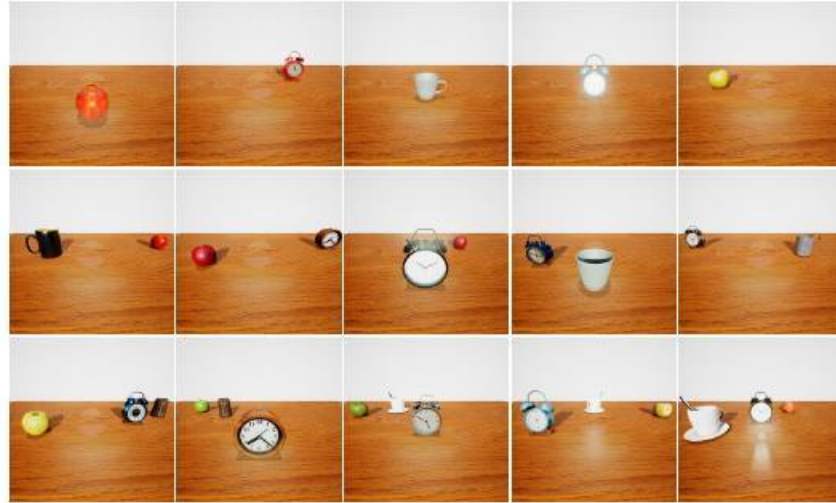
Despite this challenge, Table A3 shows that JDM can still reduce counting hallucinations in a large pretrained diffusion model under carefully controlled finetuning. Because the RealHand scale is limited, directly aligning a six-channel JDM latent space with the rigid latent geometry of a pretrained model is difficult. We therefore use a two-stage comparison: first finetune Stable-Diffusion-1.5 on RealHand, then freeze all intermediate UNet layers and finetune only the extended-channel input and output layers. Under this constrained setup, joint-diffusion finetuning lowers CHR at 25 and 50 sampling steps for both first- and second-order solvers, while the table also makes clear that NCFR and FID increase relative to the baseline.

Overall, the quantitative evidence suggests that JDM is a promising direction for count-based factual consistency, but not yet a complete replacement for perceptual-quality optimization. **Exploring its**

972 **behavior and performance under true large-scale training remains an important and compelling**
973 **avenue for future research.**
974
975
976
977
978
979
980
981
982
983
984
985
986
987
988
989
990
991
992
993
994
995
996
997
998
999
1000
1001
1002
1003
1004
1005
1006
1007
1008
1009
1010
1011
1012
1013
1014
1015
1016
1017
1018
1019
1020
1021
1022
1023
1024
1025



(a) Examples in the ToyShape dataset

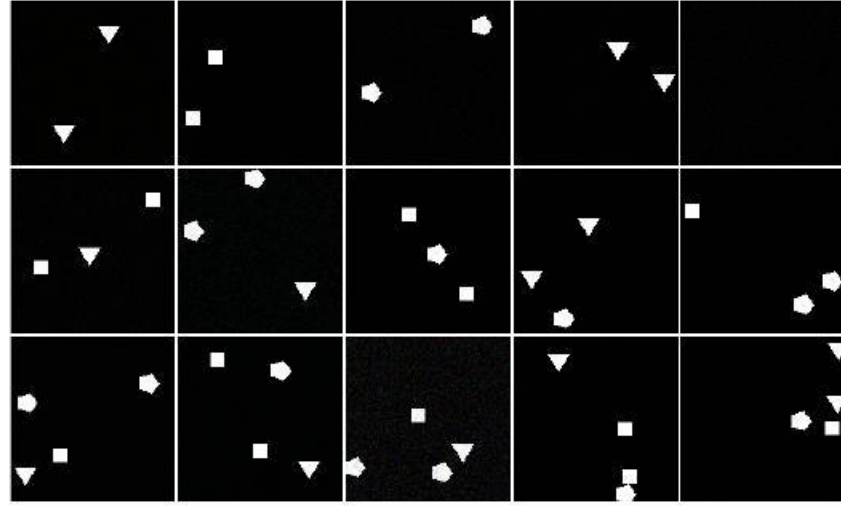


(b) Examples in the SimObject dataset



(c) Examples in the RealHand dataset

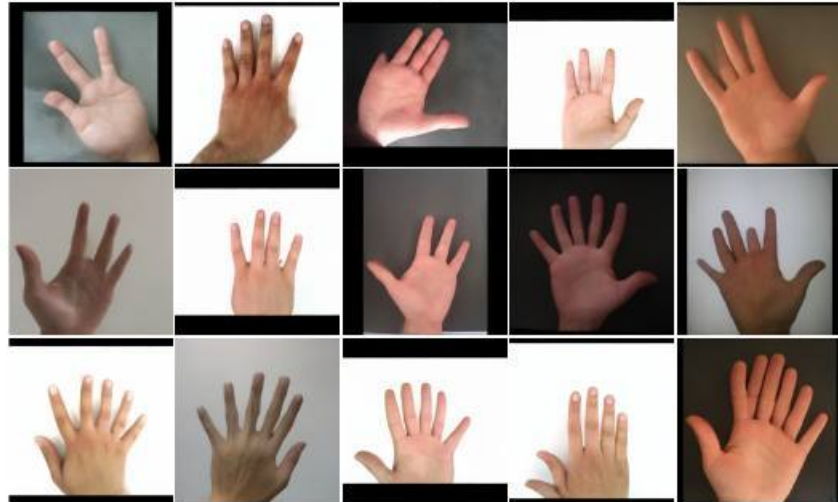
Figure A1: Examples of datasets in the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) ToyShape hallucinations: duplicate categories or empty images that violate the reference count rule

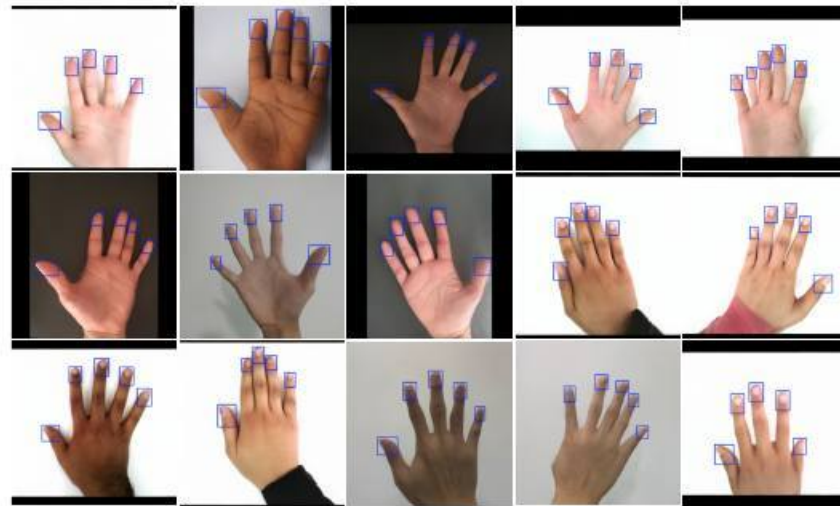


(b) SimObject hallucinations: duplicate object categories or no-object images under the tabletop rule

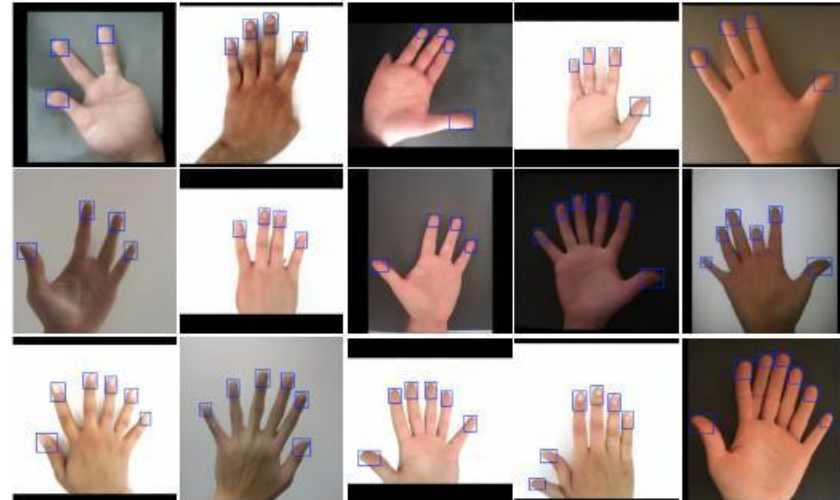


(c) RealHand hallucinations: generated hands with valid finger counts different from five

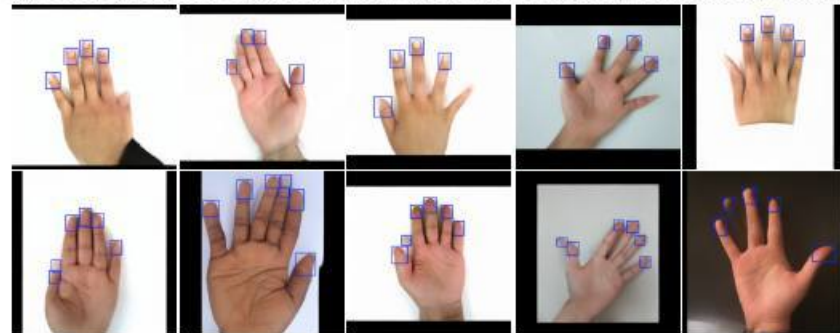
Figure A2: Dataset-specific examples of the generated samples counted as hallucinations in CHR: count-rule violations for ToyShape, SimObject, and RealHand.



(a) YOLO predictions counted as valid: five detected fingertips

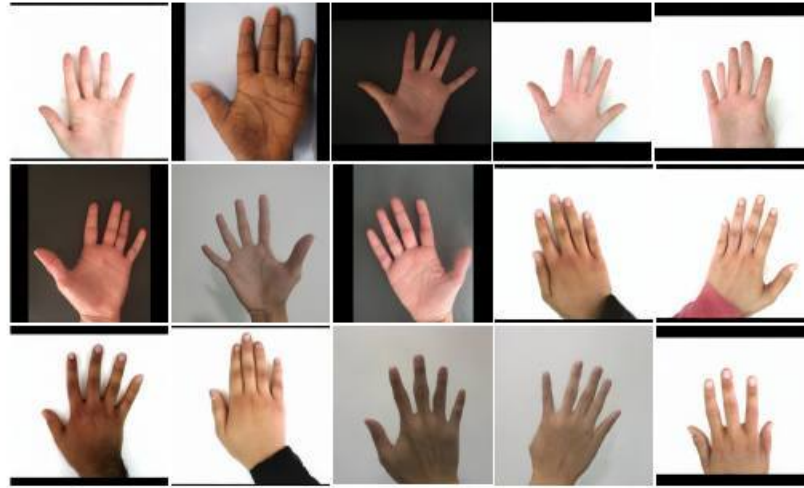


(b) YOLO predictions counted as hallucinated: detected fingertip count differs from five

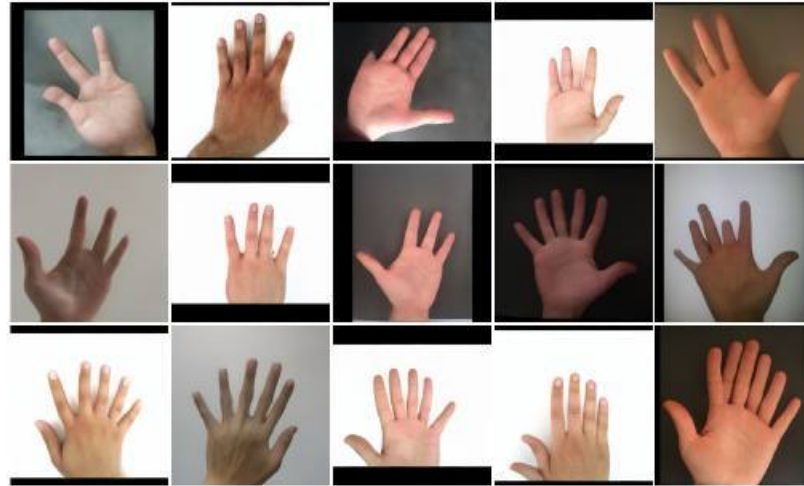


(c) Rare YOLO edge cases excluded from robust count attribution

Figure A3: YOLO fingertip predictions used for RealHand quantification. (a) Counting-valid samples have five detected fingertips, including some anatomically unusual cases. (b) Counting-hallucinated samples have non-five detections such as 3, 4, 6, or 7 fingertips. (c) Rare edge cases involve blur, incomplete fingers, or disconnected artifacts where count attribution is unreliable.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (not five valid fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A4: RealHand outcome taxonomy used to compute CHR and NCFR. (a) Counting-valid images contain five clearly discernible fingers. (b) Counting-hallucinated images are counting-ready but contain an incorrect valid finger count, such as 4 or 6. (c) Non-counting failures are unsuitable for counting because of severe artifacts, including distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (more than 5 fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A5: PTI-Hand outcome taxonomy for pretrained Stable-Diffusion-3.5-Medium. (a) Counting-valid images contain five clearly discernible fingers. (b) Counting-hallucinated images meet the conservative pretrained-model criterion of more than five valid fingers, such as 6 or 7. (c) Non-counting failures are unsuitable for counting because of severe artifacts, including distorted, cracked, partially visible, or floating fingers.